Yiming (Emmett) Peng

emmett-peng.github.io | emmett.peng@mail.utoronto.ca | Emmett Peng | Emmett-Peng

Education

- **University of Toronto** *Sept 2026 (Expected)*
  Doctor of Philosophy in Biostatistics
  Research interests: statistical machine learning, causal inference, and representation learning
  - Advisors: Kuan Liu, Zihang Lu

- **University of Toronto** *Sept 2024 – Aug 2026 (Expected)*
  Master of Science in Biostatistics – Thesis Option; GPA: 4.00/4.00
  Thesis: Bayesian Nonparametric Latent Class Framework for Causal Inference in Cognitive Trajectories
  - Advisors: Kuan Liu, Zihang Lu

- **Beijing Institute of Technology, Zhuhai** *Sept 2020 – Jun 2024*
  Bachelor of Science in Applied Statistics; GPA: 3.96/4.00 (WES)
  Thesis: Deep Learning and Stochastic Modeling of Human Activity Patterns through Smartwatch Sensors for Health Interventions (*Honored as Outstanding Thesis*)
  - Advisor: Han C.W. Hsiao

Publications and Manuscripts

R=Under Review, S=In Submission, W=Working Papers

[R] Denier, D., **Peng, Y. (Emmett)**, . . . , Ricciuto, A. (2026) **Longitudinal GGT Trajectories Identify Prognostic Phenotypes in Pediatric Primary Sclerosing Cholangitis**. *Under review at JHEP Reports*.

[S] **Peng, Y. (Emmett)***, Truong, V.*, Xie, A., Shi, Y., Hu, P. (2026) **PRECISE: A Framework for the Prediction of Response Using Cell-Type Inference and Single-Cell Embedding in Breast Cancer Patients Receiving Anti–PD-1 Treatment**. *In Submission*. **[Code]**

[W] **Peng, Y. (Emmett)**, Lu, Z., Liu, K. (2026) **Bayesian Nonparametric Latent Class Framework for Causal Inference in Cognitive Trajectories**. *Manuscript in preparation*.

[W] Zhang, J., Shi, Y., Xie, A., **Peng, Y. (Emmett)**, Hu, P. (2026) **Predicting New-Onset Atrial Fibrillation with Competing Risks Using an Uncertainty-Calibrated Interpretable Tabular Transformer: Evidence from the Cardiovascular Imaging Registry of Calgary (CIROC)**. *Manuscript in preparation*.

[W] Khan, A., **Peng, Y. (Emmett)**, Zhang, I., Chen, H. (2026) **Robust and Interpretable Machine Learning for Atypical Parkinson's Disease Classification Under Distribution Shift**. *Working paper*.

[W] Jing, Y., **Peng, Y. (Emmett)**, Danguecan, A., Knight, A. (2026) **Examining the Relationship Between Socioeconomic Factors and Mental Health in Childhood-Onset Systemic Lupus Erythematosus Using a Multilevel Analysis**. *Working paper*.

Presentations

[1] **Peng, Y. (Emmett)**, Lu, Z., Liu, K. (2026) **Bayesian Nonparametric Latent Class Framework for Causal Inference in Cognitive Trajectories**. Accepted presentations at the *Statistical Society of Canada (SSC) Annual Meeting*, *Joint Statistical Meetings (JSM)*, and *ICSA–Canada Chapter Symposium*, 2026. *Forthcoming*.

[2] **Peng, Y. (Emmett)**, Denier, D., . . . , Ricciuto, A. (2026) **Longitudinal GGT Trajectories Identify Prognostic Phenotypes in Pediatric Primary Sclerosing Cholangitis**. Poster presentation, *Canadian Statistics Student Conference (CSSC)*, Hamilton, Ontario, Canada, May 2026. *Forthcoming*.

[3] **Peng, Y. (Emmett)**, Truong, V., Xie, A., Shi, Y., Hu, P. (2025) **Enhancing Breast Cancer Treatment Response Prediction with Single-Cell RNA Sequencing and Large Language Models**. Oral presentation and poster, *2025 Statistical Society of Canada (SSC) Annual Meeting*, p. 58. May 25–28, Saskatoon, Saskatchewan, Canada. **[Slides] [Poster]**

[4] Zhang, J., Shi, Y., Xie, A., **Peng, Y. (Emmett)**, Hu, P. (2025) **Uncertainty-Calibrated Interpretable Tabular Transformer Model for Atrial Fibrillation Prediction with Competing Risk**. Poster, *2025 Statistical Society of Canada (SSC) Annual Meeting*, p. 42. May 25–28, Saskatoon, Saskatchewan, Canada. **[Poster]**

[5] **Peng, Y. (Emmett)**, Lu, Z., Liu, K. (2025) **Bayesian Latent Class Causal Inference with Nonparametric Covariate Modeling for Endpoint Cognitive Outcomes in Aging Populations**. Invited poster presentation, Data Science Institute Causal Inference Workshop, Toronto, Ontario, Canada, October 22, 2025. **[Poster]**

PROFESSIONAL EXPERIENCE

- **Knight Lab, The Hospital for Sick Children** — *Jun 2025 - Present*
  Clinical Research Project Assistant, supervised by ***Dr. Andrea Knight*** — Toronto, Canada
  - Implement Bayesian statistical methods in R to study socioeconomic and mental health outcomes in childhood-onset systemic lupus erythematosus (cSLE).
  - Lead statistical analyses, including data pre-processing, imputation, and modeling, collaborate on study design, and provide statistical consulting while contributing to manuscript preparation.
- **The Hospital for Sick Children** — *Jun 2025 - Present*
  Research Volunteer, supervised by ***Dr. Amanda Ricciuto and Dr. Anne Griffiths*** — Toronto, Canada
  - Implement latent class mixed models in R to analyze longitudinal serum biomarker trajectories in primary sclerosing cholangitis (PSC).
  - Integrate latent class membership into survival analysis to assess prediction of transplant outcomes, and collaborate with clinicians and biostatisticians on interpretation and manuscript preparation.
- **Hu Lab, Dalla Lana School of Public Health, University of Toronto** — *Oct 2024 - Jun 2025*
  Data Science Research Student, supervised by ***Dr. Pingzhao Hu*** — Toronto, Canada
  - Developed an integrated multi-stage pipeline in Python combining single-cell RNA-seq data with embeddings from large language models (LLMs) and foundation models to identify cell type-specific biomarkers for breast cancer treatment response.
  - Implemented supervised classification using cell-type–specific biomarkers and cluster-specific bulk gene-expression profiles to predict treatment response, achieving improved predictive performance over baseline.
- **The University of Hong Kong** — *Jan 2025 - Apr 2025*
  Research Volunteer, supervised by ***Dr. Liwu Zheng***
  - Collaborate with clinicians on biostatistical research projects.
  - Provide statistical consulting for clinical studies.

HONORS AND AWARDS

Amounts in Canadian dollars unless otherwise specified

- **ICSA-Canada Chapter Symposium Student/Postdoc Travel Scholarship Award - $600** — *2026*
  ICSA-Canada Chapter Symposium
- **Winner – Case Studies in Data Analysis Competition, SSC 2025 - $2,500 per team** — *2025*
  Statistical Society of Canada (SSC)
- **CSSC 2025 Travel Award - $150** — *2025*
  Canadian Statistics Student Conference (CSSC), sponsored by CANSSI
- **SGS Conference Grant - $450** — *2025*
  University of Toronto
- **BITZH President's Scholarship - $2,000** — *2023*
  Beijing Institute of Technology, Zhuhai
- **College Dean's Honor Scholarship - $7,000** — *2021 - 2024*
  Beijing Institute of Technology, Zhuhai
- **BITZH Freshman Entrance Scholarship and Continuing Scholarship - $18,000** — *2020 - 2024*
  Beijing Institute of Technology, Zhuhai
- **BITZH Outstanding Student Scholarship** — *2020 - 2024*
  Beijing Institute of Technology, Zhuhai

SKILLS

- **Programming Languages:** R, Python, MATLAB, C++
- **Tools & Technologies:** Git, Linux, HPC (Compute Canada, Slurm), LaTeX, RMarkdown, Jupyter Notebook
- **Statistical Methods:** Causal Inference, Latent Class Analysis, Survival Analysis, Longitudinal Data Analysis
- **Machine Learning Models:** Transformers, RNNs, Regression (GLM, Penalization), CNNs, Neural Networks
- **Languages:** Chinese (native), English (professional working proficiency)

COMMUNITY INVOLVEMENT AND LEADERSHIP EXPERIENCE

- **ASA DataFest at UofT** — *May 2026*
  Judge and Mentor
- **Biostatistics Union of Graduate Students (BUGS)** — *Sep 2024 - Present*
  PHSA MSc Representative and Seminar Committee member
- **Health Data Working Group** — *Sep 2024 - Present*
  Student Member
- **Public Health Students' Association, University of Toronto** — *Sep 2025 - Present*
  Biostatistics MSc Representative
- **Student Union, College of Global Talent, Beijing Institute of Technology, Zhuhai** — *Sep 2020 – Jun 2022*
  Officer